Analysis of the Chaotic Regime in the Filtered Hénon Map: Effect of FIR Filter Type and Design Window

Kauê de Souza Silva, Roger Freitas Pereira, Marcos Felipe Correa Soares, Guilherme Rodrigues dos Santos, and Vinícius da Silva Borges

***Abstract*—The insertion of finite impulse response (FIR) filters in the feedback loop of chaotic generators is a known strategy for bandwidth limitation in chaos-based communication systems (CBCS). Although the filter does not prevent master–slave synchronization, its interaction with the nonlinearity of the generating map may destroy the chaotic behavior. This work investigates, by numerical simulation, the conditions under which chaos is preserved in the Hénon map with an internal FIR filter, for four filter types (lowpass, highpass, bandpass, and bandstop) and seven design windows, including a sweep of the Kaiser shape parameter. Points of the plane formed by the filter order and the cutoff frequency whose orbits become unbounded are classified as divergent. The remaining points are classified as chaotic when the largest Lyapunov exponent, which measures the average exponential rate at which nearby orbits separate, exceeds a threshold of 5×10^{-3}, and as non-chaotic otherwise. Filtering does not eliminate the chaotic regime but reorganizes it in a way that depends on both the filter type and the window. Filters that remove the direct current (DC) component make the fixed point of the map stable and yield no chaotic point in the grid, whereas, for unit DC gain, the order, the cutoff frequency, and the window decide the regime. At the orders studied, the designed filters do not realize the nominal bandpass and bandstop masks, and the chaotic generators confine their spectrum only partially. Therefore, neither the dynamics nor the spectrum of a filtered generator follows from the filter specification, and both must be verified for each design.**

***Index Terms*—Chaos-based communication systems, Hénon map, FIR filters, window design, Lyapunov exponent.**

I. Introduction

DETERMINISTIC discrete-time chaotic signals are bounded in amplitude, aperiodic, and exhibit sensitive dependence on initial conditions (SDIC), so that orbits starting arbitrarily close separate, on average exponentially fast [1]. They also have a broadband spectrum and a noise-like appearance, which motivates their use as carriers in communication systems [2]. Among the advantages usually attributed to such carriers are the difficulty of interception by unauthorized receivers, which places them among the physical-layer security approaches [3], the spreading of the transmitted spectrum, and robustness to multipath fading in some modulation schemes [4]. This interest grew after Pecora and Carroll [5] showed that two identical chaotic systems can be synchronized despite SDIC, when the receiver is driven by state variables of the transmitter.

In chaos-based communication systems (CBCS), the message is embedded in a chaotic carrier at the transmitter and recovered at the receiver. In coherent CBCS, the receiver synchronizes a replica of the transmitter generator with the received signal and uses it to extract the message. Non-coherent architectures, in contrast, recover the information from statistical or structural properties of the received signal without a synchronized replica [4]. Examples of CBCS include chaotic modulation schemes [2], image coding and encryption [6], systems based on three-dimensional attractors [7], and phase-locked loops for systems based on chaotic attractors [8].

The broadband spectrum of chaotic signals, however, conflicts with the fact that practical channels are band-limited. If the spectrum of the transmitted signal is not confined to the channel passband, the distortion introduced by the channel prevents the receiver from synchronizing, and the message is lost [9]. To overcome this limitation, Eisencraft *et al.* [10] proposed inserting a discrete-time linear filter in the feedback loop of the chaotic generator, and Fontes and Eisencraft [11] showed that the filter does not prevent synchronization, a result already indicated in [10]. Over band-limited channels, a filter in the feedback loop of the generator allows coherent detection [10], [11]. Further practical limitations, such as finite-precision digital implementation, were investigated by Borges and Eisencraft [12]. The lowpass case corresponds to a baseband channel, but other spectral masks arise in practice. Highpass filtering removes the direct current (DC) component of the chaotic signal, which carries no information and is blocked by capacitively coupled channels. Bandpass filtering confines the signal to an allocated band, and bandstop filtering creates a spectral notch to avoid narrowband interference or to coexist with other systems [13]. From a dynamical standpoint, these four types also have different gains at zero frequency, which determine the fixed point of the filtered map [14].

K. de S. Silva is an undergraduate student in Computer Engineering at the Faculdade Engenheiro Salvador Arena (FESA), Centro Educacional da Fundação Salvador Arena, Estrada dos Alvarengas, 4001, São Bernardo do Campo, SP, Brazil. ORCID: 0009-0009-1283-0996 (e-mail: kaue.moelas.com@gmail.com).

R. F. Pereira is an undergraduate student in Computer Engineering at FESA, São Bernardo do Campo, SP, Brazil. ORCID: 0009-0000-5624-7422 (e-mail: roger_freitas@live.com).

M. F. C. Soares is an undergraduate student in Computer Engineering at FESA, São Bernardo do Campo, SP, Brazil. ORCID: 0009-0009-9968-2227 (e-mail: marcosfelipe.soares2014@gmail.com).

G. R. dos Santos is an undergraduate student in Computer Engineering at FESA, São Bernardo do Campo, SP, Brazil. ORCID: 0009-0004-0281-8663 (e-mail: guilhermerodriguess130@gmail.com).

V. da S. Borges is with the Computer Engineering Department, FESA, São Bernardo do Campo, SP, Brazil. He is the academic advisor of this work. ORCID: 0000-0001-5546-6842 (e-mail: viniciussbrg@gmail.com).

Inserting a filter in the feedback loop, however, changes the dynamics of the generator, and nothing guarantees that the transmitted signal remains chaotic. Once the unfiltered pair synchronizes, the filter preserves synchronization for any choice of coefficients [15], but it may destroy the chaotic behavior of the generator. Borges and Eisencraft showed numerically that the coefficients determine whether the generated signals are chaotic, periodic, or divergent, for a two-coefficient filter [16] and for lowpass filters of increasing order [17], with bifurcation cascades, coexisting attractors, crises, and shrimp-shaped periodic islands [18]. Borges *et al.* extended this analysis to finite impulse response (FIR) filters with more coefficients and different zero placements [14] and to second-order all-pole filters [19]. The preservation of chaos must therefore be verified for each filter design. It remains open which combinations of filter type, design window, order, and cutoff frequency keep the generator chaotic, and this is the question addressed here.

Previous studies varied the coefficients of a two-coefficient filter [16], the zero placement of the FIR filter [14], and the order and cutoff frequency of lowpass filters designed with the Hamming and Blackman windows [14], [17]. In practice, however, an FIR filter is specified by its frequency response and designed by a method such as windowing, so the filter type and the design window are natural design choices. This work investigates how these two choices affect the preservation of chaos in the Hénon map with an internal FIR filter. Lowpass, highpass, bandpass, and bandstop filters are designed with seven windows, including a sweep of the Kaiser shape parameter [20], and for each configuration the filter order and the cutoff frequency are varied to build maps that classify the generated signals as chaotic, non-chaotic, or divergent. The parameters of the Hénon map and the bandwidth of the bandpass and bandstop filters are kept fixed.

The remainder of this paper is organized as follows. Section II presents the preliminary concepts: the Hénon map, the Hénon map with an FIR filter in the feedback loop, the largest Lyapunov exponent, and the FIR design windows. Section III describes the methodology, Section IV presents and discusses the results, and Section V summarizes the conclusions, the contributions of the work, and future directions.

II. Preliminary Concepts

This section introduces the concepts needed to follow the rest of the paper. The Hénon map is presented first and then extended with an FIR filter in the feedback loop. The largest Lyapunov exponent, used to classify the signals generated by both systems, comes next, followed by the windows used to design the filter.

A. Hénon Map

Consider the Hénon map [21]

$$\begin{cases} x_1(n+1) = \alpha - x_1^2(n) + \beta\, x_2(n), \\ x_2(n+1) = x_1(n), \end{cases} \tag{1}$$

where $n \in \{0, 1, 2, \ldots\}$ is the discrete time, $x_1(n)$ and $x_2(n)$ are the state variables, and α and β are real parameters. Grouping the state variables in the vector $\mathbf{x}(n) = [x_1(n)\ \ x_2(n)]^T$, where $(\cdot)^T$ denotes transposition, (1) can be written compactly as $\mathbf{x}(n+1) = \mathbf{F}(\mathbf{x}(n))$. The sequence of states $\{\mathbf{x}(n)\}_{n\geq 0}$ obtained by iterating $\mathbf{F}$ from an initial condition $\mathbf{x}(0)$ is called an *orbit* [1].

Depending on the parameters and on the initial condition, an orbit may converge to a fixed point, converge to a periodic orbit, behave chaotically, or diverge [1]. A *fixed point* is a state $\mathbf{x}^\star$ left unchanged by one iteration, $\mathbf{x}^\star = \mathbf{F}(\mathbf{x}^\star)$, so that an orbit started at $\mathbf{x}^\star$ stays there for all n. An orbit is *bounded* when there is a finite M such that $\|\mathbf{x}(n)\| \leq M$ for all n, and *divergent* otherwise [1]. For (1), the condition $\mathbf{x}^\star = \mathbf{F}(\mathbf{x}^\star)$ gives two fixed points, of which the positive one is used as the reference state in this work.

Fig. 1 illustrates the chaotic and the periodic behaviors with segments of $x_1(n)$ generated from two initial conditions that differ by 10^{-3} in $x_2(0)$, as in [17]. In Fig. 1(a), with $\alpha = 1.4$ and $\beta = 0.3$, the signal is aperiodic and, although the two initial conditions are nearly indistinguishable, the distance between them first exceeds 0.1 at $n = 11$ and reaches order one at $n = 18$; this is the SDIC of chaotic signals. In Fig. 1(b), with $\alpha = 0.9$ and $\beta = 0.3$, both orbits converge to the same period-2 orbit, which alternates between $x_1 \approx 1.08$ and $x_1 \approx -0.38$, and the difference between the initial conditions is lost. Throughout this work, $\alpha = 1.4$ and $\beta = 0.3$ are adopted.

B. Hénon Map With an FIR Filter in the Feedback Loop

This subsection defines the filter, the expanded system obtained when the filter is inserted in the map, and the quantities of that system used in the rest of the paper. To limit the bandwidth of the generated signal, Eisencraft *et al.* [10] and Fontes and Eisencraft [11] proposed filtering $x_1(n)$ with an FIR filter and feeding the filter output back into the map. Fig. 2 shows the generator studied in this work. The output of the filter is

$$x_3(n) = \sum_{j=0}^{N_s-1} c_j\, x_1(n-j), \tag{2}$$

where c_j, $0 \leq j \leq N_s - 1$, are the N_s filter coefficients. The filter order is $N_z = N_s - 1$, equal to the number of zeros of the transfer function obtained by taking the z-transform of (2) with zero initial conditions [22],

$$H(z) = \frac{X_3(z)}{X_1(z)} = \sum_{j=0}^{N_s-1} c_j\, z^{-j}. \tag{3}$$

Evaluating $H(z)$ on the unit circle, $z = e^{j\omega}$, gives the frequency response $H(e^{j\omega})$, where $\omega \in [0, \pi]$ is the normalized angular frequency in rad/sample. The frequency response fixes the passband, the transition width, and the stopband attenuation delivered by a given set of coefficients [22]. In this work, it is specified by the filter type (lowpass, highpass, bandpass, or bandstop) and by a cutoff frequency ω_c. Its value at $\omega = 0$ is the DC gain of the filter, $G = H(e^{j0}) = \sum_j c_j$ [22].

The filtered signal $x_3(n)$ replaces $x_1(n)$ in the nonlinear term of (1). Because (2) depends on the past samples $x_1(n-1), \ldots, x_1(n-N_s+1)$, the pair (x_1, x_2) is no longer a complete state of the system. These samples are therefore stored as auxiliary state variables $x_4, \ldots, x_{N_s}$, each holding a

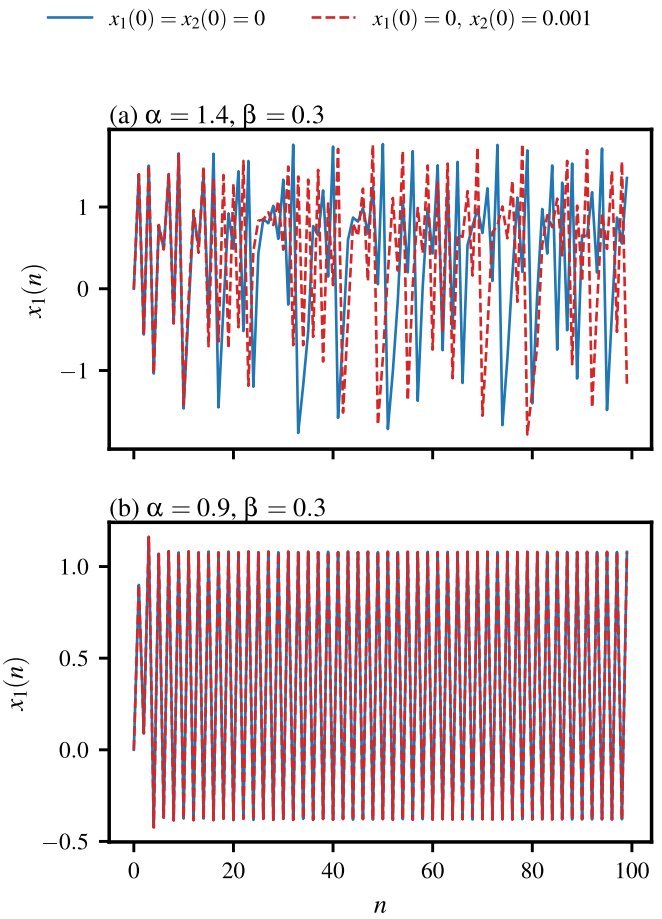


Fig. 1: Segments of $x_1(n)$ obtained from (1) with $x_1(0) = x_2(0) = 0$ (solid blue) and $x_1(0) = 0$, $x_2(0) = 0.001$ (dashed red). (a) $\alpha = 1.4$, $\beta = 0.3$: chaotic signal. (b) $\alpha = 0.9$, $\beta = 0.3$: periodic signal (period 2).

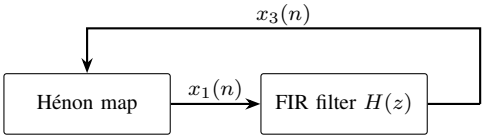


Fig. 2: Generator studied in this work, described by (4).

one-step delay of the preceding one. For $N_s \geq 3$, the resulting expanded system is [14], [17]

$$\begin{cases} x_1(n+1) = \alpha - x_3^2(n) + \beta\, x_2(n), \\ x_2(n+1) = x_1(n), \\ x_3(n+1) = c_0\big(\alpha - x_3^2(n) + \beta\, x_2(n)\big) + c_1\, x_1(n) \\ \qquad\qquad + c_2\, x_2(n) + c_3\, x_4(n) + \cdots + c_{N_s-1}\, x_{N_s}(n), \\ x_4(n+1) = x_2(n), \\ x_j(n+1) = x_{j-1}(n), \quad j = 5, \ldots, N_s, \end{cases} \tag{4}$$

whose dimension is $K = N_s$; for $N_s = 3$, the terms in $x_4, \ldots, x_{N_s}$ are absent. For $N_s \leq 2$, the filter output $x_3(n) = c_0 x_1(n) + c_1 x_2(n)$ (with $c_1 = 0$ for $N_s = 1$) is a function of the current state, so no auxiliary variable is needed and the system keeps dimension $K = 2$:

$$\begin{cases} x_1(n+1) = \alpha - \big(c_0 x_1(n) + c_1 x_2(n)\big)^2 + \beta\, x_2(n), \\ x_2(n+1) = x_1(n). \end{cases} \tag{5}$$

For $N_s = 1$ and $c_0 = 1$, the filter is the identity and the original map (1) is recovered. As for (1), the fixed points of (4) are the solutions of $\mathbf{x}^\star = \mathbf{F}(\mathbf{x}^\star)$, where $\mathbf{F}$ now denotes the right-hand side of (4). At a fixed point all delayed samples are equal, $x_1 = x_2 = x_4 = \cdots = x_{N_s} = p$, and $x_3 = Gp$, which gives $G^2 p^2 + (1-\beta)p - \alpha = 0$ [14], [15]. Its positive root is

$$p = \frac{2\alpha}{(1-\beta) + \sqrt{(1-\beta)^2 + 4\alpha G^2}}. \tag{6}$$

For $G = 0$, approximately the case of highpass and bandpass filters, $p = \alpha/(1-\beta)$. The positive fixed point of the expanded system is $\mathbf{x}^\star = [p\ \ p\ \ Gp\ \ p \cdots p]^T$ (or $[p\ \ p]^T$ for $N_s \leq 2$); it depends on the filter only through its DC gain G and is used to initialize the simulations (Section III-B).

C. Largest Lyapunov Exponent

The sensitivity of a system to its initial conditions can be quantified by how fast two very close states move apart as they evolve. Lyapunov proposed measuring this separation under the assumption that it grows exponentially, so that the Lyapunov exponent is the natural logarithm of the average rate at which two nearby orbits separate [1], [17]. An exponent equal to $\ln 2$ along a given direction, for instance, means that the distance between the orbits doubles, on average, at each iteration in that direction, whereas a negative exponent means that the orbits approach each other. The SDIC of a chaotic signal is therefore measured by the Lyapunov exponent: a computationally generated signal that is bounded, aperiodic, and has at least one positive Lyapunov exponent is chaotic [1].

A map of dimension K has K Lyapunov exponents, one for each direction of the state space, denoted in decreasing order by $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_K$. They are computed from the *Jacobian matrix* $\mathbf{J}(n)$, the $K \times K$ matrix of partial derivatives $[\mathbf{J}(n)]_{ik} = \partial F_i / \partial x_k$ evaluated at $\mathbf{x}(n)$, which describes how a small perturbation of the state is transformed in one iteration [1]. For the expanded system (4), it is [14], [17]

$$\mathbf{J}(n) = \begin{pmatrix} 0 & \beta & -2x_3(n) & 0 & \cdots & 0 \\ 1 & 0 & 0 & 0 & \cdots & 0 \\ c_1 & c_0\beta + c_2 & -2c_0 x_3(n) & c_3 & \cdots & c_{N_s-1} \\ 0 & 1 & 0 & 0 & \cdots & 0 \\ \vdots & & & \ddots & & \vdots \\ 0 & \cdots & 0 & 0 & 1 & 0 \end{pmatrix}, \tag{7}$$

in which only the third row depends on the filter coefficients; for $N_s \leq 2$, $\mathbf{J}(n)$ is the 2×2 Jacobian of (5). For the map (1), $\det \mathbf{J}(n) = -\beta$ for every n, so its two exponents satisfy $\lambda_1 + \lambda_2 = \ln|\beta|$ [1], an identity used in Section IV-A to validate the implementation.

The exponents are computed by the tangent-map method [1], [17]. A hypersphere of unit radius centered at $\mathbf{x}(0)$ is represented by a matrix $\mathbf{U}$ whose columns are its orthogonal axes. Each iteration of the map turns the

hypersphere into an ellipsoid, and after n iterations the axes of the ellipsoid are the columns of $\mathbf{J}(n-1)\cdots\mathbf{J}(1)\mathbf{J}(0)\mathbf{U}$. Computing the lengths of these axes directly is numerically inadequate, because directions that are stretched and directions that are compressed produce very large and very small numbers. Instead, the growth of each axis is computed at every iteration, and the axes are orthonormalized again before the next one, in the following steps [1]:

1) Choose an orthonormal basis $\{\mathbf{w}_1(0),\ldots,\mathbf{w}_K(0)\}$ and compute $\mathbf{z}_k(0) = \mathbf{J}(0)\mathbf{w}_k(0)$, $1 \le k \le K$. These vectors are axes of the new ellipsoid, but they are not necessarily orthogonal.
2) Orthogonalize them by the Gram–Schmidt process,

$$\mathbf{y}_k(1) = \mathbf{z}_k(0) - \sum_{i=1}^{k-1} \frac{\mathbf{z}_k(0)\cdot\mathbf{y}_i(1)}{\|\mathbf{y}_i(1)\|^2}\,\mathbf{y}_i(1), \qquad (8)$$

where "$\cdot$" denotes the scalar product.
3) Normalize the vectors, $\mathbf{w}_k(1) = \mathbf{y}_k(1)/\|\mathbf{y}_k(1)\|$, which gives the orthonormal basis for the next iteration.
4) Apply the Jacobian at the next point of the orbit, $\mathbf{z}_k(1) = \mathbf{J}(1)\mathbf{w}_k(1)$, and repeat steps 2 and 3. After n repetitions, the vectors $\mathbf{y}_k(1),\ldots,\mathbf{y}_k(n)$ are available.

The Lyapunov exponent in the k-th direction is then

$$\lambda_k = \lim_{n\to\infty} \frac{\ln\|\mathbf{y}_k(1)\| + \cdots + \ln\|\mathbf{y}_k(n)\|}{n}. \qquad (9)$$

Only the largest exponent, λ_1, is used in this work. For $\alpha = 1.4$ and $\beta = 0.3$, the map (1) has $\lambda_1 \approx 0.42$ [23].

Based on these concepts, each orbit is classified as

$$\begin{cases} \text{divergent}, & \text{if the orbit is unbounded,} \\ \text{chaotic}, & \text{if it is bounded and } \lambda_1 > \theta, \\ \text{non-chaotic}, & \text{if it is bounded and } \lambda_1 \le \theta, \end{cases} \qquad (10)$$

where $\theta = 5\times10^{-3}$ is the classification threshold. The non-chaotic class also includes convergence to a fixed point, because the criterion distinguishes only the presence or absence of chaos and not the period of the limit orbit. The numerical detection of unbounded orbits is described in Section III-C.

D. FIR Design Windows

This subsection describes how the coefficients c_j are obtained and which design windows are compared. The coefficients are obtained by the window method [22], [24]: the impulse response of the ideal filter, which is infinitely long, is conveniently delayed and truncated by a window of N_s samples. The shape of the window sets a trade-off between the stopband attenuation and the width of the transition band of the resulting filter: windows with lower side lobes give higher attenuation at the cost of a wider transition band [24].

Seven windows are considered (Table I). The values listed are the peak side-lobe levels of the discrete-time Fourier transform of each window, relative to the main-lobe peak [24]; in the window method, lower side lobes give higher stopband attenuation of the designed filter [22]. The values range from -13 dB for the rectangular window, whose abrupt truncation produces strong side lobes, to -92 dB for the four-term Blackman–Harris window, whose cosine-sum coefficients are chosen to cancel side lobes [24].

TABLE I: FIR Windows Evaluated and Their Peak Side-Lobe Levels

Window	Side lobe (dB)
Rectangular (boxcar)	-13
Bartlett	-27
Hann	-32
Hamming	-43
Blackman	-58
Blackman–Harris	-92
Kaiser (β_K variable)	variable

Source: adapted from [24].

The seven windows cover a broad range of this trade-off: the rectangular window as the worst case; the Bartlett and Hann windows as classical intermediate cases; the Hamming window, adopted as a reference because it was used in the previous works on the filtered Hénon map [11], [17]; and the Blackman and Blackman–Harris windows as high-attenuation cases. The Kaiser window [20] has a shape parameter $\beta_K \ge 0$ that varies the trade-off continuously without changing the filter order: for $\beta_K \in [0.5, 20]$ its peak side-lobe level, computed here for a window of 101 samples, goes from about -13.6 dB to -155 dB, a range that contains the side-lobe levels of the Bartlett, Hann, Hamming, Blackman, and Blackman–Harris windows in Table I, and for $\beta_K = 0$ the window reduces to the rectangular one. The sweep of β_K therefore provides a continuous path between the fixed windows. Values $\beta_K > 12$ correspond to stopband attenuations above about 120 dB [22], beyond practical requirements; they are included to locate the maximum of the chaotic coverage (Section IV-E).

III. Methodology

The research adopts a quantitative-experimental approach: for each filter design, the expanded system (4) is simulated, λ_1 is estimated, and each grid point is classified by (10). All simulations were implemented in the open-source package `chaotic_pfc` (BSD-3-Clause license) [25]; the 184 sweeps took about 10 min on a four-core laptop.

A. Filter Types and Parameter Grid

Four filter types are designed with the seven windows of Table I by the routine `firwin` of the SciPy library [26]. For lowpass and highpass filters, ω_c is the cutoff frequency; for bandpass and bandstop filters, it is the center of a band of fixed total width $\Delta\omega = 0.05\,\pi$, placed symmetrically around ω_c. Near the ends of the grid, the band edges are clipped to the interval $(0, 1)$ of ω/π, so the band is narrower and asymmetric for the cutoff values closest to 0 and to 1. On the normalized axis $\omega/\pi \in (0, 1)$, this width is 5% of the range and spans about five steps of the cutoff grid. This $\Delta\omega = 0.05\,\pi$ is nominal: at the orders of the grid, the realized filter is much wider for bandpass and barely rejects the band for bandstop (for a band centered at $\omega_c = 0.5\pi$, with the six fixed windows at $N_z = 10$, the realized bandpass 3 dB width is between $0.18\,\pi$ and $0.38\,\pi$ and the bandstop attenuation at the

band center between 0.70 and 2.04 dB; at $N_z = 40$, only the rectangular window among the fixed windows reaches 20 dB, whereas the others stay below 6.2 dB). Fixing $\Delta\omega$ also isolates the effect of ω_c; the width is left as a variable for future work (Section V).

Because it is the frequency response, and not the individual coefficients, that must be matched to the channel, the sweep variables are the filter order N_z and the cutoff frequency ω_c. For lowpass and bandpass filters, $N_z \in \{1, 2, \ldots, 40\}$ and 100 values of ω_c/π linearly spaced from 10^{-5} to $1 - 10^{-5}$ are swept, giving 4,000 grid points per window–filter configuration. For highpass and bandstop filters, the parity restriction of the window method requires even N_z [22], because a filter with an odd order has a forced zero at $\omega = \pi$, which a passband that includes $\omega = \pi$ does not allow; thus $N_z \in \{2, 4, \ldots, 40\}$, with 2,000 points per configuration. The six fixed windows give $4 \times 6 = 24$ configurations, and the Kaiser window is swept over 40 values of $\beta_K \in [0.5, 20.0]$, with step 0.5, for the four filter types, adding 160 configurations, for a total of 184 sweeps and 552,000 grid points. The case $\beta_K = 0$, which coincides with the rectangular window, was also swept as a check and reproduced the rectangular configurations exactly; it is not counted separately. For the Hann and Bartlett windows with lowpass and bandpass filters, `firwin` returns non-finite coefficients for $N_z = 1$, because the symmetric two-sample windows are zero; these 100 points per configuration are classified as divergent because of the filter design, not of the map dynamics.

B. Lyapunov Estimator and Initial Conditions

Computing all K exponents by the method of Section II-C costs $\mathcal{O}(K^3)$ operations per iteration, dominated by the Gram–Schmidt orthogonalization. As in [14], λ_1 is taken as the largest of the K finite-time exponents. To reduce the cost, the sweeps first propagate a single tangent vector, multiplied by $\mathbf{J}(n)$ at each iteration and renormalized once every ten iterations, which costs $\mathcal{O}(K)$ per iteration (Algorithm 1); the formulation follows Wolf *et al.* [27], and the renormalization period follows Benettin *et al.* [28], [29]. Grid points whose mean estimate lies within 5×10^{-3} of the threshold θ of (10), that is, $|\lambda_1 - \theta| < 5 \times 10^{-3}$, are recomputed with the full spectrum, using the modified Gram–Schmidt algorithm; 23,000 of the 552,000 grid points (4.17%) were recomputed. The first direction of the full-spectrum computation follows the same path as the single tangent vector, so the recomputation changes the estimate only when another direction has a larger finite-time exponent.

Accumulation stops early when the estimate of λ_1 varies less than the tolerance $\varepsilon = 10^{-3}$ in two consecutive checks, made every 100 iterations after the 500th. Requiring two checks avoids premature stops in windows of chaotic intermittency, in which λ_1 oscillates for hundreds of iterations before stabilizing.

For each grid point (N_z, ω_c), $N_{\text{IC}} = 25$ initial conditions are drawn near the positive fixed point $\mathbf{x}^\star$ of the expanded system (Section II-B): each component is $x_k^\star + 0.1\,u$, with $u \sim U[0, 1)$. For each initial condition, $N_{\text{burn}} = 500$ iterations are discarded as transient, and λ_1 is accumulated for at most $N_{\text{map}} = 3{,}000$ iterations. The value assigned to the grid point is the mean of λ_1 over the initial conditions.

Algorithm 1 Single-vector estimation of λ_1 with periodic renormalization.

Input: initial condition $\mathbf{x}_0$; burn-in iterations N_{burn}; accumulation cap N_{map}; tolerance ε.
$\mathbf{w} \leftarrow [1\ 0\ \cdots\ 0]^T$; $s \leftarrow 0$; $\mathbf{x} \leftarrow \mathbf{x}_0$.
for $i = 1$ **to** N_{burn} **do**
  $\mathbf{x} \leftarrow \mathbf{F}(\mathbf{x})$ (discard transient)
  if $\mathbf{x}$ is non-finite **then**
    return DIVERGENT (grid point)
  end if
end for
for $i = 1$ **to** N_{map} **do**
  $\mathbf{w} \leftarrow \mathbf{J}(\mathbf{x})\,\mathbf{w}$ (tangent-vector evolution at the current state)
  $\mathbf{x} \leftarrow \mathbf{F}(\mathbf{x})$ (orbit evolution)
  if $\mathbf{x}$ is non-finite **then**
    return NaN (divergent orbit)
  end if
  if i is a multiple of 10 **then**
    $r \leftarrow \|\mathbf{w}\|$; $s \leftarrow s + \ln\max(r, 10^{-300})$; if $r > 10^{-300}$ then $\mathbf{w} \leftarrow \mathbf{w}/r$
    every 100 iterations with $i \geq 500$: stop if s/i varied less than ε in two consecutive checks
  end if
end for
return $\lambda_1 = s/i$

The value $N_{\text{burn}} = 500$ is much longer than the transients of the unfiltered map and accommodates longer transients near bifurcation boundaries, although not all of them (Section IV-F). The value N_{map} acts as a cap: with early stopping, the median number of accumulated iterations per configuration lies between 700 and 1,200, with a floor of 700, and the mean horizon reaches the cap at only 6 of the 552,000 grid points. In all experiments, $\alpha = 1.4$, $\beta = 0.3$, and the seed of the pseudo-random generator is 42.

C. Classification and Chaotic-Coverage Metrics

Each grid point is classified by (10). Numerically, an orbit is taken as unbounded when its state becomes non-finite (NaN or infinity); from that iteration on, every term of (9) is non-finite, so the simulation can be stopped. The divergence rule has three steps. First, the estimator returns a non-finite value for an initial condition as soon as its state becomes non-finite. Second, a grid point is divergent when at least one of its 25 initial conditions becomes non-finite during the burn-in. Third, an initial condition that becomes non-finite only after the burn-in is discarded, and the value of the grid point is the mean of λ_1 over the remaining initial conditions. This step differs from [14], in which a grid point is divergent when any of its initial conditions becomes non-finite. Points with non-finite filter coefficients (Section III-A) are also divergent.

The threshold $\theta = 5 \times 10^{-3}$ in (10) separates two populations of finite-time estimates. Over the 184 sweeps, 13,068 of

the 54,470 grid points with a positive estimate lie in $(0, 10^{-3}]$, whereas bins of width 10^{-3} between 5×10^{-3} and 2×10^{-2} hold between 400 and 516 points each. The first population is consistent with quasi-periodic orbits, whose exponent is zero but whose finite-time estimate decays as $1/N$ with the number N of iterations: for 30 lowpass points with $N_z = 3$ and $0 < \lambda_1 \leq \theta$, recomputed with $N = 3 \times 10^3$, 3×10^4, and 3×10^5, 18 kept a positive estimate at all horizons, with a median log–log slope of -0.96, and the other 12 changed sign.

Let N_{grid} be the number of grid points of a configuration, N_b the number of bounded (chaotic or non-chaotic) points, and N_c the number of chaotic points. The *conditional* fraction $\rho_c^{\text{cond}} = N_c/N_b$ is the share of chaotic points among the bounded ones; it measures the tendency to chaos once divergence is excluded and is relevant when N_z and ω_c can be adjusted to reach a non-divergent operating point. The *absolute* fraction $\rho_c^{\text{abs}} = N_c/N_{\text{grid}}$ is the share of the whole design space that yields a chaotic generator and is relevant when a broad operating range is sought. Using fractions also makes the comparison between filter types independent of their different grid sizes (Section III-A).

D. Statistical Treatment

Two levels are kept separate. At the point level, the value of λ_1 of each grid point is the mean over its N_{IC} initial conditions. At the regional level, the chaotic region of each configuration is summarized by the mean $\bar{\lambda}_1$ and the standard deviation σ_{λ_1} of λ_1 over its N_c chaotic points; thus σ_{λ_1} measures how homogeneous λ_1 is across the chaotic region, not the estimation noise at one point. The classification is checked in three ways: the sensitivity of N_c to the threshold, with 10^{-3} and 10^{-2} in place of θ; the repetition of the best Kaiser and the best fixed window of each filter type with ten seeds of the initial-condition generator (1 to 9 and 42); and a long-horizon test, in which 400 chaotic points per filter type, drawn at random, are recomputed with 3×10^5 iterations and no early stopping from the first initial condition of their ensemble (the classification averages the 25 initial conditions). Rank correlations between window properties and N_c use the Spearman coefficient r_s over the six fixed windows, with the exact two-sided p-value over the 720 permutations.

IV. Results and Discussion

This section presents the results of the 184 sweeps.

A. Validation of the Estimator

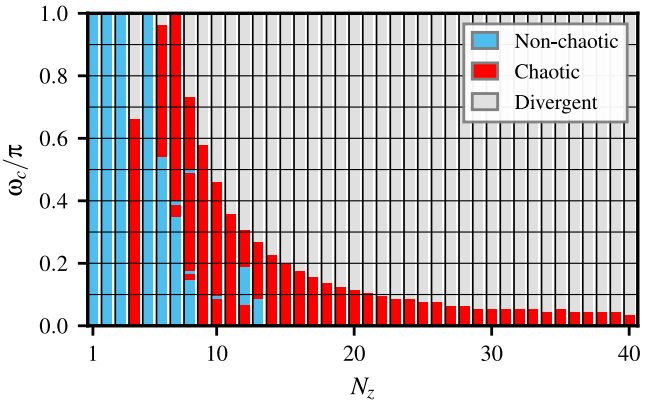
The estimator was validated on the map (1) without filter ($N_s = 1$, $c_0 = 1$), with $\alpha = 1.4$ and $\beta = 0.3$, for which $\lambda_1 = 0.4192$ [23], using the ensemble of Section III-B. The full-spectrum estimator, with early stopping disabled, gave $\bar{\lambda}_1 = 0.4185$ (0.17% below the reference), and the identity $\lambda_1 + \lambda_2 = \ln|\beta|$ (Section II-C) was satisfied within 5.4×10^{-9} (maximum over the 25 initial conditions), which confirms the estimator independently of the reference value. The single-vector path of Algorithm 1 used in the sweeps, with renormalization every ten iterations and early stopping enabled, gave $\bar{\lambda}_1 = 0.4194$, a relative difference of 0.22% from the full-spectrum value and of 0.04% from the reference, which is enough to separate the classes of (10).

B. Configurations With Largest Chaotic Coverage

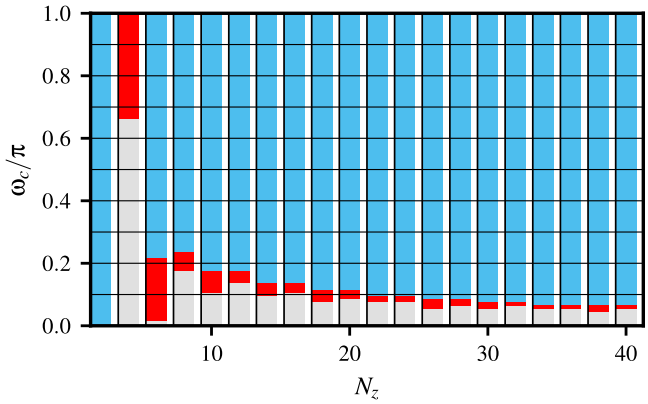
Table II ranks the six fixed windows and the best Kaiser window of each filter type by ρ_c^{abs}. The Kaiser window gives the largest N_c for the lowpass ($\beta_K = 17$, $\rho_c^{\text{abs}} = 16.4\%$), highpass ($\beta_K = 15.5$, 5.1%), and bandpass ($\beta_K = 15.5$, 7.8%) filters, and the rectangular window for the bandstop filter (5.9%). Four of the first five positions are lowpass filters, which confirms this type as the one of largest absolute chaotic coverage, whereas highpass and bandstop filters occupy mostly the lower half of the table. Repeating the best Kaiser and the best fixed window of each type with ten seeds gives, as mean $\pm$ standard deviation, $N_c = 651.1 \pm 2.6$ and 570.4 ± 2.6 (lowpass), 101.6 ± 1.3 and 73.8 ± 1.8 (highpass), 309.0 ± 0.9 and 270.4 ± 1.7 (bandpass), and, for the bandstop filter, 116.0 ± 1.8 for the rectangular window and 106.7 ± 1.7 for the Kaiser window with $\beta_K = 0.5$. The two configurations of each type differ by about 5 (bandstop) to 30 (lowpass and bandpass) standard deviations, with the same order for all ten seeds, so the ordering in Table II is not a seed effect. The comparison must still be read with caution, because each fixed window is a single configuration, whereas the Kaiser window was evaluated at 40 values of β_K and the best one was selected (Section IV-E).

Among the fixed windows, Blackman–Harris gives the largest N_c for the lowpass, highpass, and bandpass filters. Over the six fixed windows, the Spearman coefficient between the side-lobe attenuation of Table I and N_c is $r_s = +0.94$ ($p = 0.017$) for lowpass, $+0.60$ ($p = 0.24$) for highpass, $+1.00$ ($p = 0.003$) for bandpass, and -0.37 ($p = 0.50$) for bandstop filters. The side-lobe attenuation, however, is confounded with the selectivity of the designed filter: windows of lower side lobes give wider transition bands, so at the orders of the grid they also filter less. Replacing the side-lobe level by the median ratio between the realized passband width (-6 dB for lowpass and highpass, 3 dB for bandpass filters) and the nominal one, over the orders $N_z \in \{2, 4, 6, 10, 20, 40\}$ and the cutoffs $\omega_c/\pi \in \{0.1, 0.2, 0.4\}$, gives $r_s = +1.00$ ($p = 0.003$) for lowpass, $+0.94$ ($p = 0.017$) for highpass, and $+0.83$ ($p = 0.058$) for bandpass filters. The two window properties increase together across the fixed windows, and their orderings differ only among the Bartlett, Hann, and Hamming windows, where N_c follows neither of them consistently. With six windows, the data do not separate the two effects; only the coefficients of the lowpass passband width and of the bandpass side-lobe level remain significant after a Bonferroni correction over the four filter types ($0.05/4 = 0.0125$).

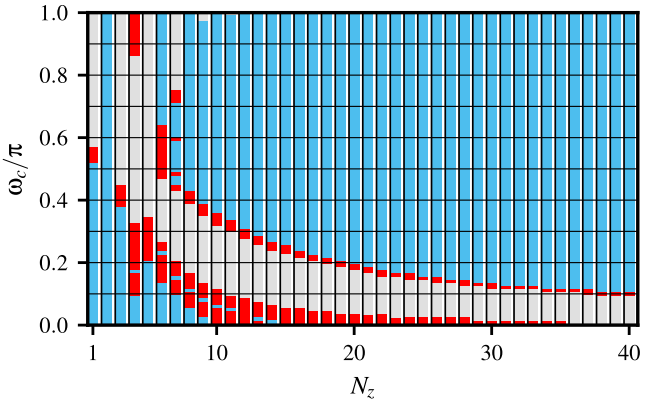
For lowpass filters designed with the Hamming and Blackman windows, Borges and Eisencraft [17] observed similar curves of λ_1 versus cutoff frequency and concluded that the window seemed to have little influence on the generation of chaos. In the present sweep, in contrast, the Blackman window gives 359 chaotic points and the Hamming window



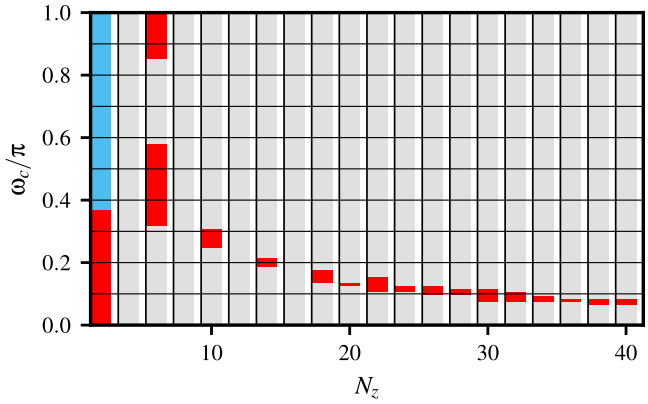
(a) Lowpass filter, Kaiser window ($\beta_K = 17$). Chaos lies below a cutoff boundary that decreases with N_z, with divergence above it.



(b) Highpass filter, Kaiser window ($\beta_K = 15.5$). Chaos occurs at $\omega_c/\pi \leq 0.23$ for $N_z \geq 6$ and at $0.67 \leq \omega_c/\pi \leq 1$ for $N_z = 4$.



(c) Bandpass filter, Kaiser window ($\beta_K = 15.5$), nominal $\Delta\omega = 0.05\,\pi$. Fragmented chaotic region, more sensitive to ω_c.



(d) Bandstop filter, rectangular window, nominal $\Delta\omega = 0.05\,\pi$. Only $N_z = 2$ has no divergent points.

Fig. 3: Classification maps in the $(N_z, \omega_c/\pi)$ plane for the configuration of largest N_c of each filter type: chaotic points in red, non-chaotic in blue, divergent in gray.

252 (Table II). The two studies differ in the grid of orders and cutoff frequencies, in the ensemble of initial conditions (10 drawn in $[0, 1]$ in [17], 25 drawn near $\mathbf{x}^\star$ here), and in the comparison criterion (curves of λ_1 there, counts of chaotic points above θ here).

The bandstop filter behaves differently. With the rectangular window it reaches $\rho_c^{\text{cond}} = 65.2\%$, the highest conditional fraction of Table II, but its absolute coverage is only 5.9%, because most of its grid diverges (Section IV-C). The rectangular window is also the only fixed window whose designed filter forms a notch at the orders of the grid (Section III-A), and the Kaiser window with $\beta_K = 0.5$, the closest to it, is the second bandstop configuration ($N_c = 108$, $\rho_c^{\text{cond}} = 62.4\%$). The other five fixed windows give between 42 and 55 chaotic points.

C. Classification Maps

Fig. 3 shows the classification maps in the $(N_z, \omega_c/\pi)$ plane for the configuration of largest N_c of each filter type. In the lowpass map (Fig. 3(a)), chaos lies below a boundary in ω_c/π that decreases with N_z, and the orbits diverge above it. The largest chaotic cutoff is 0.66 for $N_z = 4$, 0.96 for $N_z = 6$, and close to 1 for $N_z = 7$, and it falls to 0.45 for $N_z = 10$, 0.11 for $N_z = 20$, and 0.03 for $N_z = 40$; for $N_z = 3$ and 5 no point is chaotic. In the highpass map (Fig. 3(b)), chaos occurs at $\omega_c/\pi \leq 0.23$ for $N_z \geq 6$ and at $0.67 \leq \omega_c/\pi \leq 1$ only for $N_z = 4$, where the largest λ_1, 0.105, also occurs. At high orders, both filters are chaotic only at low cutoff frequencies. The bandpass map (Fig. 3(c)) shows a fragmented chaotic region, more sensitive to ω_c. In the bandstop map (Fig. 3(d)), chaos appears in isolated bands and divergence dominates the grid: only $N_z = 2$ has no divergent points, and for $N_z = 4$, 8, 12, and 16 the whole row diverges.

Pooling the 184 sweeps, chaos is limited mainly by divergence in lowpass and bandstop filters, in which 69.7% and 92.6% of the grid points diverge, and mainly by non-chaotic orbits in highpass and bandpass filters, in which 89.0% and 82.3% of the points are non-chaotic.

This grouping follows the DC gain G of the filters, which sets the fixed point (6). Lowpass and bandstop filters have $G = 1$ at every grid point, whereas highpass and bandpass filters have $G \approx 0$ except at low orders and at cutoffs close to $\omega = 0$, where the designed filter does not remove the DC component. None of the 182,892 grid points with $|G| < 0.1$ is chaotic, and

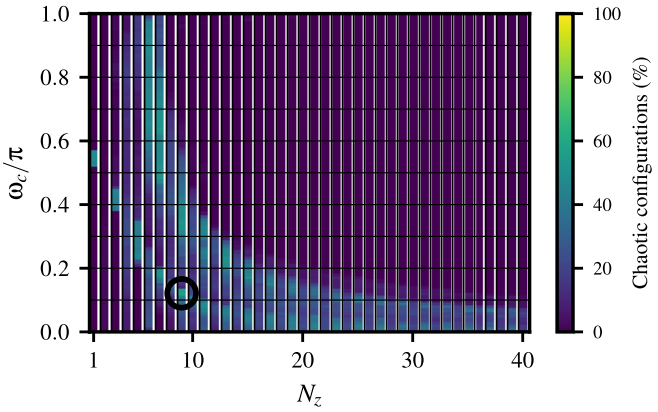


Fig. 4: Aggregate chaotic density over the 184 sweeps: percentage of the configurations containing each grid point in which it is chaotic (184 configurations for even N_z, 92 for odd N_z, which have only lowpass and bandpass filters). The marker (○) indicates the maximum, 56.5% at $N_z = 9$ and $\omega_c/\pi \approx 0.12$.

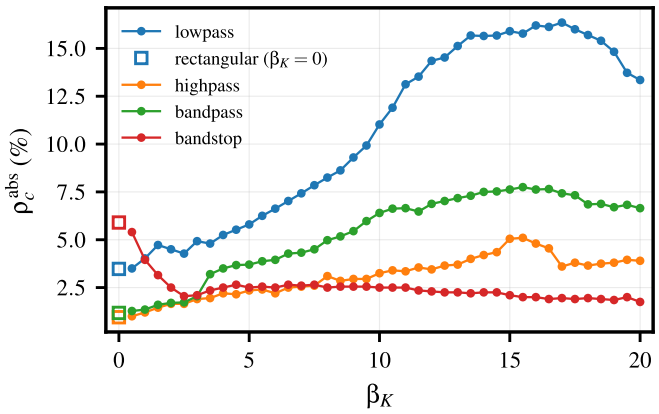


Fig. 5: Absolute chaotic coverage ρ_c^{abs} versus the Kaiser shape parameter β_K for the four filter types. The square marks the rectangular window ($\beta_K = 0$).

all 2,790 chaotic points of the highpass filter have $|G| \geq 0.1$. The reason is visible in the Jacobian (7): for $G = 0$, the fixed point has $x_3^\star = Gp = 0$, the terms in $-2x_3$ vanish, the block of (x_1, x_2) decouples with eigenvalues $\pm\sqrt{\beta} \approx \pm 0.548$, and the remaining eigenvalues are zero. The fixed point $\mathbf{x}^\star$, with $p = \alpha/(1-\beta)$, is then linearly stable for any choice of coefficients, and orbits started near it converge to it, so the generated signal is constant. Numerically, all but 41 of the 182,795 bounded grid points with $|G| < 0.1$ converge to $\mathbf{x}^\star$ within 10^{-8}, and drawing the 25 initial conditions uniformly in $[-1.5, 1.5]^K$ instead of near $\mathbf{x}^\star$ changed the class of none of 200 such points chosen at random. The non-chaotic class therefore consists almost only of the fixed point when $|G| < 0.1$, whereas for $|G| \geq 0.1$ the share of non-chaotic points that converge to $\mathbf{x}^\star$ is 21%, 34%, 45%, and 0% for lowpass, highpass, bandpass, and bandstop filters, respectively. A unit DC gain, in contrast, does not determine the class: the lowpass filter, with $G = 1$ at every point with finite coefficients, has 10.3% chaotic, 20.0% non-chaotic, and 69.7% divergent points, so the order, the cutoff, and the window decide the class once the DC component is preserved.

D. Aggregate Chaotic Density

To identify the regions of the $(N_z, \omega_c/\pi)$ plane that are chaotic regardless of the design, the chaotic points of the 184 sweeps were aggregated. Because highpass and bandstop filters have only even N_z, a grid point with even N_z belongs to 184 configurations and one with odd N_z to 92 (lowpass and bandpass only). Fig. 4 therefore shows, at each grid point, the percentage of the configurations containing it in which it is chaotic.

For $N_z \geq 8$, chaos is most frequent at low cutoff frequencies: on average, a grid point is chaotic in 23.7% of the configurations for $\omega_c/\pi \leq 0.1$, in 12.6% for $0.1 < \omega_c/\pi \leq 0.2$, in 4.9% for $0.2 < \omega_c/\pi \leq 0.4$, and in 0.7% above $\omega_c/\pi = 0.4$. For $N_z \leq 7$ the trend is reversed, and the same averages are 4.8%, 9.8%, 11.7%, and 15.4%. The maximum, 56.5%, occurs at $N_z = 9$ and $\omega_c/\pi \approx 0.12$ (52 of 92 configurations); among even orders, it occurs at $N_z = 6$ and $\omega_c/\pi \approx 0.89$ (90 of 184, 48.9%). Odd and even orders are not directly comparable, because the odd ones include only the lowpass and bandpass filters, which have more chaotic points; this explains the alternation between neighboring columns in Fig. 4.

The choice of window matters mainly at low and moderate orders. Excluding the grid points whose filter coefficients are not finite, the six fixed windows give the same class at 61.6% of the lowpass grid points with $N_z \leq 10$ and at 88.9% of those with $N_z > 20$; the corresponding values are 62.9% and 89.1% for bandpass, 83.4% and 95.0% for highpass, and 79.4% and 97.4% for bandstop filters. This convergence is expected, because the designed filters approach the ideal one as N_z grows [22].

E. Effect of the Kaiser Parameter β_K

Fig. 5 shows ρ_c^{abs} versus β_K. For the lowpass filter, ρ_c^{abs} grows from 3.5% at $\beta_K = 0.5$ and 4.5% at $\beta_K = 2$ to 15.1% at $\beta_K = 13$, stays between 15.6% and 16.4% up to $\beta_K = 17.5$, and falls to 13.4% at $\beta_K = 20$. For the bandpass filter, it grows from 1.3% at $\beta_K = 0.5$ to values between 7.2% and 7.8% for $13 \leq \beta_K \leq 17.5$ and falls to 6.7% at $\beta_K = 20$. For the highpass filter, it grows from 1.0% at $\beta_K = 0.5$ to a maximum of 5.1% at $\beta_K = 15.5$ and stays between 3.6% and 4.0% for $\beta_K \geq 17$. The maxima, at $\beta_K = 17$ (lowpass) and 15.5 (highpass and bandpass), are therefore interior to the sweep and, for the lowpass filter, one value inside a plateau: ten seeds give $N_c = 651.1 \pm 2.6$ at $\beta_K = 17$ and 645.0 ± 3.3 at $\beta_K = 16$. Beyond the maxima, the coverage decreases while the side-lobe attenuation of the window keeps increasing and its main lobe keeps widening, so the coverage is not a monotonic function of either property, which agrees with the confounding discussed in Section IV-B.

For the bandstop filter, the coverage is highest at the smallest β_K: 5.4% at $\beta_K = 0.5$, just below the 5.9% of the rectangular window ($\beta_K = 0$), falling to 2.5% at $\beta_K = 2$ and staying between 1.8% and 2.7% for larger β_K. Over the same range,

TABLE II: Ranking of the Window–Filter Combinations (Fixed Windows and Best Kaiser per Filter Type), by ρ_c^{abs}

Rank	Filter	Window	N_c	ρ_c^{abs} (%)	ρ_c^{cond} (%)	$\bar{\lambda}_1$	σ_{λ_1}
1	Lowpass	Kaiser ($\beta_K = 17.0$)	654	16.4	55.3	0.0423	0.0186
2	Lowpass	Blackman–Harris	572	14.3	44.5	0.0347	0.0211
3	Lowpass	Blackman	359	9.0	27.7	0.0381	0.0227
4	Bandpass	Kaiser ($\beta_K = 15.5$)	310	7.8	9.5	0.0435	0.0272
5	Lowpass	Hann	273	6.8	23.1	0.0411	0.0223
6	Bandpass	Blackman–Harris	271	6.8	8.0	0.0388	0.0293
7	Lowpass	Hamming	252	6.3	20.1	0.0466	0.0229
8	Bandstop	Rectangular	118	5.9	65.2	0.0707	0.0492
9	Bandstop	Kaiser ($\beta_K = 0.5$)	108	5.4	62.4	0.0729	0.0433
10	Bandpass	Blackman	213	5.3	6.0	0.0457	0.0348
11	Highpass	Kaiser ($\beta_K = 15.5$)	102	5.1	5.7	0.0406	0.0284
12	Lowpass	Bartlett	194	4.9	17.5	0.0454	0.0246
13	Bandpass	Hamming	167	4.2	4.5	0.0481	0.0333
14	Bandpass	Hann	155	3.9	4.3	0.0487	0.0336
15	Highpass	Blackman–Harris	75	3.8	4.1	0.0352	0.0292
16	Bandpass	Bartlett	139	3.5	3.8	0.0518	0.0341
17	Lowpass	Rectangular	139	3.5	13.3	0.0515	0.0270
18	Highpass	Bartlett	56	2.8	3.0	0.0356	0.0285
19	Bandstop	Hamming	55	2.8	35.5	0.0870	0.0071
20	Highpass	Blackman	55	2.8	3.0	0.0371	0.0287
21	Bandstop	Hann	51	2.6	33.8	0.0879	0.0064
22	Bandstop	Blackman	50	2.5	33.3	0.0888	0.0086
23	Highpass	Hann	49	2.5	2.6	0.0366	0.0290
24	Bandstop	Blackman–Harris	48	2.4	32.4	0.0933	0.0062
25	Highpass	Hamming	47	2.4	2.5	0.0350	0.0288
26	Bandstop	Bartlett	42	2.1	29.6	0.0890	0.0057
27	Bandpass	Rectangular	47	1.2	1.2	0.0441	0.0390
28	Highpass	Rectangular	19	1.0	1.0	0.0347	0.0302

Full per-configuration tables are regenerated from the versioned sweep files by the released package [25].

the attenuation at the center of the band ($N_z = 40, \omega_c = 0.5\pi$) falls from 20.1 dB for $\beta_K = 0$ to 18.7, 15.8, 13.2, and 11.2 dB for $\beta_K = 0.5$, 1.0, 1.5, and 2.0, so the bandstop coverage decreases together with the depth of the realized notch. The effect of β_K is therefore not the same for the four filter types; it depends on the spectral mask imposed.

F. Robustness of the Classification

Three checks support the classification. First, N_c depends little on the threshold above θ: replacing θ by 10^{-2} reduces the N_c of the best configuration of each type by 2.9% (lowpass), 15.7% (highpass), 2.3% (bandpass), and 2.5% (bandstop), whereas lowering it to 10^{-3} raises the lowpass count by 17%, in agreement with the population of marginal estimates discussed in Section III-C. Second, the ten-seed repetitions of Section IV-B give standard deviations of N_c of at most 3.3 points. Third, in the long-horizon test, the share of sampled chaotic points that keep $\lambda_1 > \theta$ after 3×10^5 iterations is 89.0% (95% CI 85.6 to 91.7) for lowpass, 89.5% (86.1 to 92.1) for bandpass, 83.3% (79.3 to 86.6) for highpass, and 69.8% (65.1 to 74.0) for bandstop. The points that end in a non-chaotic orbit or diverge, that is, chaotic transients longer than the horizon of the sweeps, are 3.0% (1.7 to 5.2), 3.0% (1.7 to 5.2), 11.3% (8.5 to 14.7), and 26.0% (21.9 to 30.5), respectively; the remaining ones keep a positive estimate below θ. The lowpass and bandpass classifications are therefore the most robust, whereas a measurable fraction of the highpass and bandstop chaotic points is transient.

The dispersion σ_{λ_1} is highest for the rectangular bandstop filter (0.049), whose fragmented chaotic region mixes low and high values of λ_1. The bandstop configurations also have the highest $\bar{\lambda}_1$ (up to 0.093), but they have the largest share of transients in the long-horizon test (26.0%, 95% CI 21.9 to 30.5), so this value should not be read as a property of a sustained chaotic attractor.

G. Spectrum of the Generated Signals

The filter is inserted to confine the spectrum of the transmitted signal $x_3(n)$, so the chaotic configurations are also evaluated by their spectra. Fig. 6 shows, for each filter type, the normalized Welch PSD of $x_3(n)$ and $|H(e^{j\omega})|^2$ at a representative chaotic point of the configuration of largest N_c: the one whose λ_1 is closest to the median of the chaotic region, among the points with $0.125 \le \omega_c/\pi \le 0.875$. To quantify the confinement, let f be the fraction of the power of $x_3(n)$ inside the nominal passband ($[0, \omega_c]$ for lowpass, $[\omega_c, \pi]$ for highpass, the band of width $\Delta\omega$ for bandpass, and its complement for bandstop filters), and f_0 the same fraction for $x_1(n)$ of the unfiltered map (1). The ratio f/f_0 measures how much the filtered generator concentrates its power in the nominal passband relative to the unfiltered one. Over the chaotic points of the four configurations, computed with 2^{16} samples, the median of f/f_0 is 10.1 for the lowpass filter with $N_z \ge 8$ (10th and 90th percentiles 2.6 and 43), but only 1.6 for $N_z \le 7$. For the bandpass filter, it is 5.8 for $N_z \ge 8$ and 0.8 for $N_z \le 7$, and most of the power remains outside the nominal band (median $f = 0.04$). For the highpass and bandstop filters, f/f_0 is close to 1 (medians 1.00 and 1.02), because, at the cutoffs where most of their chaotic points occur, their nominal passbands cover nearly the whole frequency axis. In highpass and bandstop generators, chaotic signals can also carry strong spectral lines: the five largest peaks of the PSD of $x_3(n)$ hold

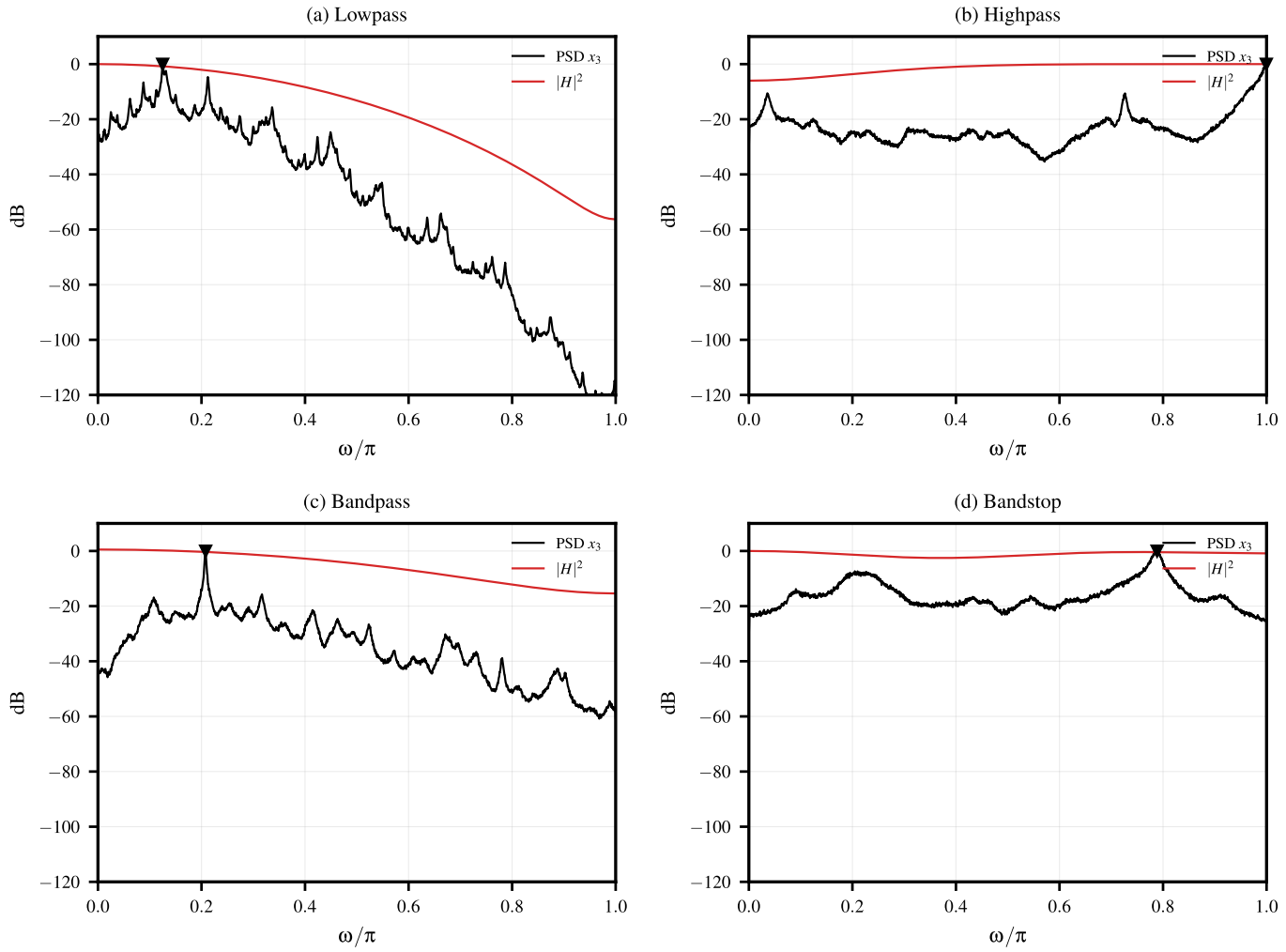


Fig. 6: Normalized Welch PSD of $x_3(n)$ (black) and $|H(e^{j\omega})|^2$ of the filter (red) at a representative chaotic point of the configuration of largest N_c of each filter type (the point whose λ_1 is closest to the median of the chaotic region, with $0.125 \leq \omega_c/\pi \leq 0.875$): (a) lowpass, (b) highpass, (c) bandpass, (d) bandstop. (a) $N_z = 10$, $\omega_c/\pi = 0.192$. (b) $N_z = 12$, $\omega_c/\pi = 0.141$. (c) $N_z = 6$, $\omega_c/\pi = 0.162$. (d) $N_z = 6$, $\omega_c/\pi = 0.394$.

a median of 15% and 5% of the power, and up to 34% and 66% at the 90th percentile, respectively; the points shown in Fig. 6(b) and (d) hold 7% and 4%.

These results connect with Section IV-C. At high orders, chaos occurs only at low cutoffs, where the lowpass filter does concentrate the power, by a median factor of 10, in a narrow band; at low orders, chaos extends to high cutoffs, but the filter changes the in-band power little. Every chaotic bandpass point has $|G| \geq 0.1$, that is, a designed filter that still passes part of the DC component. The chaotic point of Fig. 6(c), at $N_z = 6$, has $G \approx 1.06$: at this order, the designed bandpass filter passes the DC component entirely, and the highpass point of Fig. 6(b) still passes half of it ($G \approx 0.50$). A chaotic generator therefore does not, by itself, realize the spectral confinement for which the filter was inserted, and this confinement must be verified together with the dynamics.

V. Final Remarks

This work investigated numerically how the filter type and the design window affect the preservation of chaos in the Hénon map with an FIR filter in the feedback loop. Four filter types (lowpass, highpass, bandpass, and bandstop) and seven windows (rectangular, Bartlett, Hann, Hamming, Blackman, Blackman–Harris, and Kaiser, with a sweep of its shape parameter β_K) were combined in 184 sweeps of the plane formed by the filter order and the cutoff frequency, and each point was classified as chaotic, non-chaotic, or divergent by the largest Lyapunov exponent with a threshold $\theta = 5 \times 10^{-3}$.

Filtering does not eliminate the chaotic regime but reorganizes it into regions whose shape depends on the filter type and the window. The DC gain of the filter decides whether chaos is possible: a filter that removes the DC component makes the fixed point of the map stable for any choice of coefficients, and no grid point with $|G| < 0.1$ was chaotic. Once the DC component is preserved, the order, the cutoff frequency, and the window decide the regime. For lowpass, highpass, and

bandpass filters, windows of high side-lobe attenuation (Kaiser with β_K between 15.5 and 17, and Blackman–Harris) give the largest chaotic coverage, the highest being that of the lowpass filter with the Kaiser window. At the orders studied, however, side-lobe attenuation and filter selectivity vary together, the data do not separate their effects, and the coverage has an interior maximum in β_K. The window matters mainly at low and moderate orders: the six fixed windows disagree on the class of 38% of the lowpass grid points with $N_z \leq 10$.

The work makes three contributions. First, it extends the previous studies of the filtered Hénon map [14], [16], [17], in which the filters designed by windowing were lowpass filters with the Hamming or the Blackman window, to four filter types and seven design windows, with classification maps that show where chaos is preserved in each case, and it explains the absence of chaos for filters that remove the DC component. Second, it separates sustained chaos from marginal estimates with a threshold supported by the distribution of finite-time exponents, and checks the classification with ten seeds and a long-horizon test. Third, it releases an open-source package [25] with which all the maps and tables can be reproduced and the study extended.

For the design of chaotic generators for CBCS, the results point in two directions. The spectral specification of the filter does not ensure a chaotic generator, and a chaotic generator does not ensure the spectral specification: at the orders studied, the designed bandpass and bandstop filters do not realize their nominal band, and the chaotic generators concentrate their power in the nominal passband only partially. Both the dynamics and the spectrum of each design must therefore be verified explicitly, for example with classification maps and power fractions such as the ones presented here.

The study has limitations. The classification uses a threshold on a finite-time estimate of λ_1: weakly chaotic orbits with $0 < \lambda_1 \leq \theta$ are counted as non-chaotic, and chaotic transients longer than the horizon of the sweeps are counted as chaotic, which affects 26.0% (95% CI 21.9 to 30.5) of the bandstop chaotic points in the long-horizon test. The initial conditions are drawn near the fixed point, so the classification is local; it was checked against a wider region only for points with $|G| < 0.1$. The value of a grid point is the mean of λ_1 over its initial conditions, so coexisting attractors are merged into a single class, and initial conditions that diverge after the burn-in are excluded from that mean. At low orders, the designed filters do not meet their nominal specification, so the filter type names the design target rather than the realized response. Finally, the sweeps are limited to $N_z \leq 40$, $N_{\text{map}} \leq 3{,}000$, a fixed band width $\Delta\omega = 0.05\pi$, and the map parameters $(\alpha, \beta) = (1.4, 0.3)$; other regions of the parameter space of the Hénon map [18] may behave differently under filtering.

Three directions are proposed for future work: (a) separating the effects of side-lobe attenuation and selectivity with design methods that control them independently, such as equiripple design [22]; (b) sweeping the band width $\Delta\omega$ of bandpass and bandstop filters at orders high enough to realize it; and (c) extending the methodology to infinite impulse response (IIR) filters, whose dynamic behavior is still little explored in the literature [19], with explicit treatment of the stability of the filter itself.

REFERENCES

[1] K. T. Alligood, T. D. Sauer, and J. A. Yorke, *Chaos: An Introduction to Dynamical Systems*, ser. Textbooks in Mathematical Sciences. New York, NY, USA: Springer-Verlag New York, 1996.

[2] M. S. Baptista, "Chaos for communication," *Nonlinear Dynamics*, vol. 105, no. 2, pp. 1821–1841, 2021.

[3] A. Mukherjee, S. A. A. Fakoorian, J. Huang, and A. L. Swindlehurst, "Principles of physical layer security in multiuser wireless networks: A survey," *IEEE Communications Surveys & Tutorials*, vol. 16, no. 3, pp. 1550–1573, 2014.

[4] G. Kaddoum, "Wireless chaos-based communication systems: A comprehensive survey," *IEEE Access*, vol. 4, pp. 2621–2648, 2016.

[5] L. M. Pecora and T. L. Carroll, "Synchronization in chaotic systems," *Physical Review Letters*, vol. 64, no. 8, pp. 821–824, 1990.

[6] E. G. Sabatini, M. T. M. Silva, and M. Eisencraft, "Codificação de imagens usando sinais caóticos," in *Anais do XXXVIII Simpósio Brasileiro de Telecomunicações e Processamento de Sinais (SBrT)*, 2020.

[7] C. E. C. Souza, D. P. B. Chaves, and C. Pimentel, "Digital communication systems based on three-dimensional chaotic attractors," *IEEE Access*, vol. 7, pp. 10 523–10 532, 2019.

[8] R. V. Silva, D. P. B. Chaves, C. Pimentel, and M. S. Baptista, "Uma nova malha de travamento de fase aplicada a sistemas de comunicação baseados em atratores caóticos," in *Anais do XLIII Simpósio Brasileiro de Telecomunicações e Processamento de Sinais (SBrT)*, Natal, RN, Brazil, 2025.

[9] M. Eisencraft, R. D. Fanganiello, and L. H. A. Monteiro, "Chaotic synchronization in discrete-time systems connected by bandlimited channels," *IEEE Communications Letters*, vol. 15, no. 6, pp. 671–673, 2011.

[10] M. Eisencraft, R. D. Fanganiello, and L. A. Baccalá, "Synchronization of discrete-time chaotic systems in bandlimited channels," *Mathematical Problems in Engineering*, vol. 2009, pp. 1–12, 2009.

[11] R. T. Fontes and M. Eisencraft, "A digital bandlimited chaos-based communication system," *Communications in Nonlinear Science and Numerical Simulation*, vol. 37, pp. 374–385, 2016.

[12] V. S. Borges and M. Eisencraft, "O efeito da digitalização num sistema de comunicação baseado em sincronismo caótico," in *Anais do XXXIX Simpósio Brasileiro de Telecomunicações e Processamento de Sinais (SBrT)*, Virtual, 2021.

[13] H.-B. Li, K. Yata, K. Takizawa, N. Miyazaki, T. Okada, K. Ohno, T. Mochizuki, E. Nakagawa, and T. Kobayashi, "Examination of effective UWB avoidance based on experiments for coexistence with other wireless systems," *IEICE Transactions on Fundamentals of Electronics, Communications and Computer Sciences*, vol. E96-A, no. 1, pp. 274–284, 2013.

[14] V. S. Borges, M. T. M. Silva, and M. Eisencraft, "Chaotic properties of an FIR filtered Hénon map," *Communications in Nonlinear Science and Numerical Simulation*, vol. 131, p. 107845, 2024.

[15] V. S. Borges, "Dinâmica de sistemas de tempo discreto compostos por mapas não lineares e filtros lineares," Ph.D. dissertation, Escola Politécnica, Universidade de São Paulo, São Paulo, Brazil, 2026.

[16] V. S. Borges and M. Eisencraft, "A filtered Hénon map," *Chaos, Solitons & Fractals*, vol. 165, p. 112865, 2022.

[17] ——, "Um estudo numérico do expoente de Lyapunov do mapa de Hénon com filtro FIR," in *Anais do XL Simpósio Brasileiro de Telecomunicações e Processamento de Sinais (SBrT)*, Sta. Rita do Sapucaí, MG, Brazil, 2022.

[18] J. A. C. Gallas, "Structure of the parameter space of the Hénon map," *Physical Review Letters*, vol. 70, no. 18, pp. 2714–2717, 1993.

[19] V. S. Borges, M. T. M. Silva, and M. Eisencraft, "Um estudo numérico da dinâmica do mapa de Hénon com filtros só-polos de segunda ordem," in *Anais do XLII Simpósio Brasileiro de Telecomunicações e Processamento de Sinais (SBrT)*, Belém, PA, Brazil, 2024.

[20] J. F. Kaiser and R. W. Schafer, "On the use of the I_0-sinh window for spectrum analysis," *IEEE Transactions on Acoustics, Speech, and Signal Processing*, vol. 28, no. 1, pp. 105–107, 1980.

[21] M. Hénon, "A two-dimensional mapping with a strange attractor," *Communications in Mathematical Physics*, vol. 50, no. 1, pp. 69–77, 1976.

[22] A. V. Oppenheim and R. W. Schafer, *Discrete-Time Signal Processing*, 3rd ed. Upper Saddle River, NJ, USA: Prentice Hall, 2009.

[23] J. C. Sprott, *Chaos and Time-Series Analysis*. Oxford, UK: Oxford University Press, 2003.

[24] F. J. Harris, "On the use of windows for harmonic analysis with the discrete Fourier transform," *Proceedings of the IEEE*, vol. 66, no. 1, pp. 51–83, 1978.

[25] K. de S. Silva, R. F. Pereira, M. F. C. Soares, G. R. dos Santos, and V. S. Borges, "chaotic-pfc: Parametric characterization of the filtered Hénon map," Software, BSD-3-Clause license, 2026. [Online]. Available: https://doi.org/10.5281/zenodo.22740317

[26] P. Virtanen *et al.*, "SciPy 1.0: Fundamental algorithms for scientific computing in Python," *Nature Methods*, vol. 17, pp. 261–272, 2020.

[27] A. Wolf, J. B. Swift, H. L. Swinney, and J. A. Vastano, "Determining Lyapunov exponents from a time series," *Physica D: Nonlinear Phenomena*, vol. 16, no. 3, pp. 285–317, 1985.

[28] G. Benettin, L. Galgani, A. Giorgilli, and J.-M. Strelcyn, "Lyapunov characteristic exponents for smooth dynamical systems and for Hamiltonian systems; a method for computing all of them. Part 1: Theory," *Meccanica*, vol. 15, pp. 9–20, 1980.

[29] ——, "Lyapunov characteristic exponents for smooth dynamical systems and for Hamiltonian systems; a method for computing all of them. Part 2: Numerical application," *Meccanica*, vol. 15, pp. 21–30, 1980.